

CustomSight: Enhancing LLM-Powered Visual Assistance for Blind Individuals using Goal-Directed Dynamic Filters

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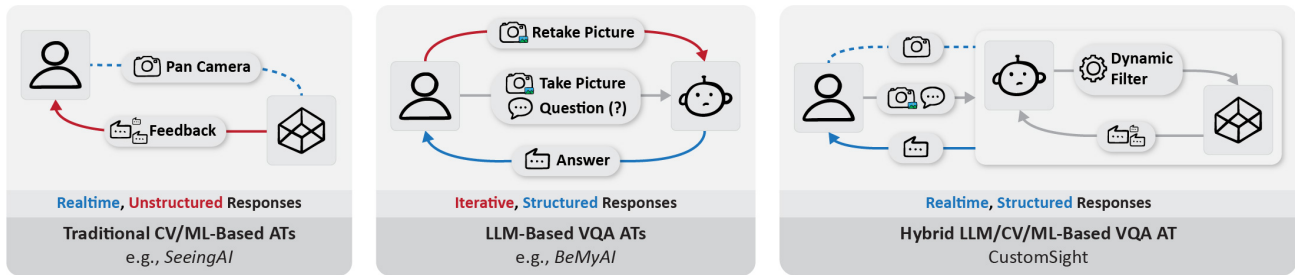


Figure 1: Computer vision-based assistive apps offer unstructured feedback, while LLM-based visual question answering apps require well-framed images, often challenging without sight. CustomSight creates Dynamic Filters to automatically capture goal-relevant images and query the LLM to assist BVI individuals in accessing targeted visual information effectively.

Abstract

LLM-powered assistive technologies (ATs) have enabled blind and visually impaired (BVI) users to query personalized, goal-oriented information about their visual environment. However, the accuracy of system responses depends heavily on well-framed, query-relevant images, which can be difficult for BVI users to capture. We present **CustomSight**, an LLM-powered AT that helps BVI users effectively query visual information by providing task-aware, real-time guidance to frame the camera and automatically capture images when relevant content is in view. When a user issues a query, CustomSight generates a *Dynamic Filter*—a custom pipeline that encodes logic tied to the user’s intent, monitors the live feed, and triggers context-aware feedback and image capture. The captured image is sent to the LLM to fetch accurate visual information.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; **Accessibility technologies**.

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Keywords

Accessibility; Blind and Low Vision; Assistive Technology; LLMs

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1 Introduction

Blind and visually impaired (BVI) individuals frequently rely on smartphone-based ATs to independently access visual information in daily life. Several computer vision (CV) models—such as text recognition, object detection, and scene captioning—have been integrated into popular ATs (e.g., *Seeing AI* [9], *Tap Tap See* [5]) to help users explore their surroundings and identify objects [6, 7]. However, despite widespread adoption, these solutions typically provide context-agnostic responses, forcing BVI users to creatively “hack” around limitations or cope with significant cognitive load to find relevant information [7].

Recently, LLM-powered ATs (e.g., *Be My AI* [2]) have introduced natural language querying, allowing BVI users to ask targeted visual questions conversationally for diverse use cases [1, 8, 10]. However, these systems depend critically on capturing relevant, well-framed

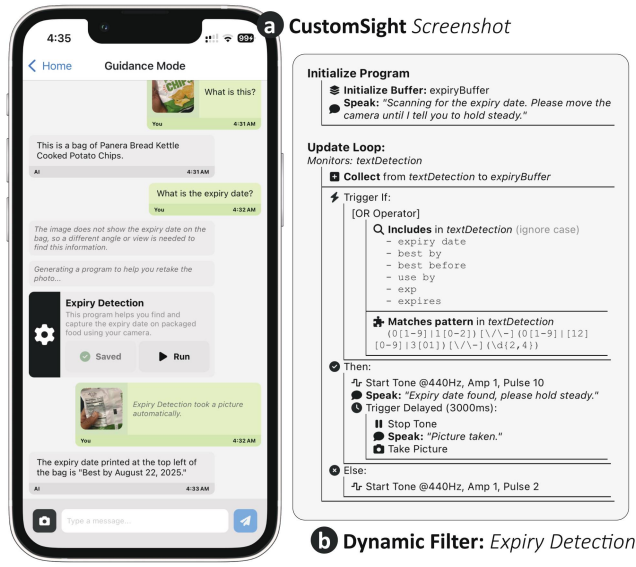


Figure 2: Finding expiry date using CustomSight. (a) CustomSight uses a chat-based interface, generating custom pipelines to support the user’s goal. (b) The dynamic filter logic is LLM-generated and defined using a custom DSL.

images to produce useful responses—an inherently difficult task without sight [3]. For instance, if a user wants to know a food item’s expiry date, the image must clearly show it. Given the asynchronous nature of LLM responses [4], capturing a usable image may take multiple attempts, reducing efficiency and limiting practicality.

To address this, we designed *CustomSight*, an assistive system that creates custom pipelines to automatically capture images relevant to a user’s goal and query the LLM, helping BVI individuals access goal-oriented visual information effectively. CustomSight retains a chat-based interface similar to existing LLM-powered apps like *Be My AI*. However, instead of simply facilitating question answering, CustomSight first checks whether the submitted image contains enough information to meaningfully answer the query. If not, it assesses whether available on-device *sensing* and *feedback* modules can help the user better aim the camera and trigger capture when relevant content is likely in frame. It then generates a **Dynamic Filter**—a tailored pipeline that processes the live feed using goal-specific logic, provides context-aware audio cues to assist framing, and automatically captures an image once the user’s goal criteria are met, passing it to the LLM for a task-specific response (Fig 2a). For example, if a user looks for expiry date, CustomSight may generate a Dynamic Filter that monitors text recognition output for relevant keywords like “expiry date”, “use by”, and “best before”, provides continuous sonification and speech feedback, and captures when relevant content is detected and steady (Fig 2b).

2 CustomSight: System Design

System Pipeline: CustomSight operates in four stages.

Step 1: Determine Answerability. When a user submits a query (e.g., image and question), CustomSight first checks whether it

can be meaningfully answered based on the current image. The system sends the user’s query and chat history, along with a rubric-based prompt, to an LLM [11], which classifies the input as either sufficient (the image contains enough information to answer the query) or retake (a better image is needed). If sufficient, the LLM responds in the chat; otherwise, CustomSight guides the user to capture a better image.

Step 2: Determine Filter Programmability. CustomSight evaluates whether on-device sensing and feedback modules can be leveraged to detect frames likely to contain relevant information. Similar to the previous step, the system provides the LLM with the user query and a list of available modules, and checks against a rubric to determine whether capture logic can be generated. If not, the LLM responds directly to the user; otherwise, the system proceeds to generate a dynamic filter.

Step 3: Generate Dynamic Filter. Dynamic Filters are task-specific routines that monitor outputs from on-device sensing modules to guide image capture. Each filter is defined using a custom domain-specific language (DSL) (Fig 2b), specifying which modules to use, how to provide feedback, and when to trigger auto-capture. If a filter is deemed programmable, the LLM generates it by inferring goal-relevant logic and stop conditions from the user’s query. For example, if a user asks for cooking instructions, the filter may monitor OCR for keywords like “instructions” or “microwave” and capture once the text is detected and stable. A short title and description are also generated to summarize the filter’s purpose.

Step 4: Execute Filter. CustomSight executes the dynamic filter on-device. As the user pans the camera, the system evaluates trigger conditions continuously and provides real-time feedback. When the condition is met, it automatically captures the frame and sends it to the LLM to answer the user’s query.

Implementation Details: CustomSight is implemented as a React Native app. It uses Apple’s VisionKit for text detection and YOLOv8 for object detection as core sensing modules. Feedback is delivered via a tone generator and TTS reader for real-time audio guidance. The DSL supports logical operations like keyword matching and pattern-based triggers. For LLM-based decision-making and program generation, we use custom-prompts on OpenAI’s GPT-4o API (gpt-4o-2024-05-13) served via a Flask/Socket.IO backend.

3 Preliminary Evaluation

We conducted an exploratory study with two blind participants (1M, 1F, ages 30 and 38) to evaluate the initial design of CustomSight. One participant was an experienced AT specialist; the other had recently lost vision and was newer to using visual ATs, offering perspectives from both expert and emerging users. Participants completed four tasks: identifying cooking instructions on a boxed item, reading the expiry date on a can, locating an orange in the room, and finding the price of a food item on a multi-page restaurant menu. Sessions lasted ~90 minutes; participants received 35 USD compensation.

Both participants had previously used LLM-powered question answering apps, but found tasks like these difficult due to framing challenges, which often disrupted the overall experience. In contrast, they found CustomSight more interactive and efficient, helping them retrieve needed information with fewer steps. Participants especially appreciated the real-time guidance and reported

higher trust in the system due to its ability to automatically capture the image, which gave the sense that the system deemed the framing to be appropriate. While the current system's binary feedback (based on inclusion or regex matching) was helpful for live scanning, it sometimes triggered too early—for example, when the word “cooking instructions” was visible at the edge of the frame but the actual content remained out of view. Participants expressed a desire for more nuanced feedback, such as audio cues to help center the target. They also envisioned future use cases, including depth-based sonification to avoid bumping into people, or smart home camera integration to detect delivered packages.

4 Conclusion & Future Work

We presented the design of CustomSight, an LLM-powered visual assistive tool that helps blind users efficiently capture task-relevant images through real-time, goal-directed guidance. Our preliminary study highlighted its potential to improve the efficiency of visual information access for BVI users. Future work includes co-design workshops to explore new sensing and feedback modules (e.g., depth, haptics), supporting more expressive framing guidance, and real-world deployments to assess everyday utility and impact.

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References

- [1] Rudaiba Adnin and Maitraye Das. 2024. “I look at it as the king of knowledge”: How Blind People Use and Understand Generative AI Tools. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility* (St. John's, NL, Canada) (*ASSETS '24*). Association for Computing Machinery, New York, NY, USA, Article 64, 14 pages. <https://doi.org/10.1145/3663548.3675631>
- [2] Be My Eyes. 2023. Introducing: Be My AI. <https://www.bemyeyes.com/blog/introducing-be-my-ai/>.
- [3] Jeffrey P. Bigham, Chandrika Jayant, Hanjie Ji, Greg Little, Andrew Miller, Robert C. Miller, Robin Miller, Aubrey Tatarowicz, Brandyn White, Samuel White, and Tom Yeh. 2010. VizWiz: Nearly Real-Time Answers to Visual Questions. In *Proceedings of the 23rd Annual ACM Symposium on User Interface Software and Technology (UIST '10)*. Association for Computing Machinery, New York, NY, USA, 333–342. <https://doi.org/10.1145/1866029.1866080>
- [4] Ruei-Che Chang, Yuxuan Liu, and Anhong Guo. 2024. WorldScribe: Towards Context-Aware Live Visual Descriptions. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24)*. Association for Computing Machinery, New York, NY, USA, 1–18. <https://doi.org/10.1145/3654777.3676375>
- [5] CloudSight, Inc. 2012. TapTapSee: Assistive Camera App for the Blind and Visually Impaired. <https://taptapseeapp.com/>.
- [6] Ricardo E Gonzalez Penuela, Jazmin Collins, Cynthia Bennett, and Shiri Azenkot. 2024. Investigating Use Cases of AI-Powered Scene Description Applications for Blind and Low Vision People. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 901, 21 pages. <https://doi.org/10.1145/3613904.3642211>
- [7] Jaylin Herskovitz, Andi Xu, Rahaf Alharbi, and Anhong Guo. 2023. Hacking, Switching, Combining: Understanding and Supporting DIY Assistive Technology Design by Blind People. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (*CHI '23*). Association for Computing Machinery, New York, NY, USA, Article 57, 17 pages. <https://doi.org/10.1145/3544548.3581249>
- [8] Florian Mathis and Johannes Schöning. 2025. LifeInsight: Design and Evaluation of an AI-Powered Assistive Wearable for Blind and Low Vision People Across Multiple Everyday Life Scenarios. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–25. <https://doi.org/10.1145/3706598.3713486>
- [9] Microsoft. 2017. Seeing AI: Talking Camera App for the Blind and Low Vision. <https://www.seeingai.com/>.
- [10] Jingyi Xie, Rui Yu, He Zhang, Syed Masum Billah, Sooyeon Lee, and John M. Carroll. 2025. Beyond Visual Perception: Insights from Smartphone Interaction of Visually Impaired Users with Large Multimodal Models. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/3706598.3714210>
- [11] Michael J. Q. Zhang and Eunsol Choi. 2023. Clarify When Necessary: Resolving Ambiguity Through Interaction with LMs. <https://doi.org/10.48550/arXiv.2311.09469> arXiv:2311.09469 [cs]